Biagio La Rosa

Curriculum Vitae

Sapienza University of Rome
(+39) 3273727639
larosa@diag.uniroma1.it
My WebPage
Linkedin

Research Interest

Techniques to **inspect** the knowledge learned by the most advanced AI techniques and **explain** their behavior.

Education

2020-2024 Apr **PhD, Engineering in Computer Science**, *Sapienza University of Rome*, Italy, Dissertation: Intrinsic Methods for eXplaining Deep Learning. Advisor: Roberto Capobianco.

2022-2023 **Visiting Scholar, Baskin School of Engineering**, *University of California, Santa Cruz*, United States, Advisor: Leilani Gilpin.

2016-2020 **Master of Artificial Intelligence and Robotics**, *Sapienza University of Rome, Italy*, Grade: 110/110, Thesis's topic: Explaining Memory Augmented Neural Networks. Advisor: Roberto Capobianco.

2010-2016 **Bachelor of Computer Science**, *Sapienza University of Rome, Italy*, Grade: 102/110, Thesis's topic: Word Sense Disambiguation. Advisor: Roberto Navigli.

Honors and Awards

2022 **Top 10% Reviewer**, *AISTATS 2022.*

2020 Three Years **Ph.D. Scholarship** from Sapienza University of Rome

2020 **"eXplainable Deep Learning" Research Scholarship** from Department of Computer, Control, and Management Engineering Antonio Ruberti

2020 **DiSCo Scholarship** from Lazio region

2016 **"SINAPSI" Scholarship** from Consorzio Interuniversitario Nazionale per l'Informatica (CINI)

2012 **DiSCo Scholarship** from Lazio region

2011 **DiSCo Scholarship** from Lazio region

Publications

2023 **[8] Biagio La Rosa, Leilani Gilpin, and Roberto Capobianco**, Towards a fuller understanding of neurons with Clustered Compositional Explanations.
In proceedings of the Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS 2023)

2023 **[7] Michela Proietti, Alessio Ragno, Biagio La Rosa, Rino Ragno, and Roberto Capobianco**, Explainable AI in Drug Discovery: Self-interpretable Graph Neural Network for molecular property prediction using Concept Whitening.
Machine Learning (Journal)

2023 **[6] Biagio La Rosa, Graziano Blasilli, Romain Bourqui, David Auber, Giuseppe Santucci, Roberto Capobianco, Enrico Bertini, Romain Giot, and Marco Angelini**, The State of The Art of Visual Analytics for eXplainable Deep Learning.
Computer Graphic Forum (Journal) and presented at EuroVIS Conference 2023

2022 **[5] Alessio Ragno, Biagio La Rosa, and Roberto Capobianco**, Prototype-based Interpretable Graph Neural Networks.
IEEE Transactions on Artificial Intelligence Journal

2022 **[4] Biagio La Rosa, Roberto Capobianco and Daniele Nardi**, A self-interpretable module for deep image classification on small data.
Applied Intelligence (Journal)

2021 **[3] Sayo M. Makinwa, Biagio La Rosa, Roberto Capobianco**, Detection Accuracy for Evaluating Compositional Explanations of Units.
In Proceedings of 20th International Conference of the Italian Association for Artificial Intelligence (AIxIA 2021)

2021 **[2] Biagio La Rosa, Roberto Capobianco and Daniele Nardi**, A discussion about Explainable Inference on Sequential Data via Memory-Tracking.
AIxIA 2021 Discussion Papers

2020 **[1] Biagio La Rosa, Roberto Capobianco and Daniele Nardi**, Explainable Inference on Sequential Data via Memory-Tracking.
In Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI-20)

Grants

2022 **Starting Research Grant**, *Sapienza University of Rome*.

Teaching

Teaching Assistant

2023 **Seminars**, *Masters in Artificial Intelligence & Robotics*, Sapienza University of Rome.

2021 **Reinforcement Learning**, *Masters in Artificial Intelligence & Robotics*, Sapienza University of Rome.

Guest Lectures and Invited Talks

2023 **Talk**, *State-of-the-Art of Visual Analytics for explainable Deep Learning*, AI, Culture & Society Club, Postdam University.

2022 **Lecture**, *eXplainable Artificial Intelligence*, Masters in Artificial Intelligence & Robotics, Sapienza University of Rome.

Mentoring

PhD Students

2022-2023 **Michela Proietti**, *Project Co-Advisor*, Co-authored paper on concept-based graph neural networks [7].

2021-2022 **Alessio Ragno**, *Project Supervisor*, Co-authored paper on prototype-based graph neural networks [5].

Master Students

2021-2022 **Sayo Makinwa**, *Thesis Supervisor*, Co-authored paper on compositional explanations [3].

2021-2022 **Giulia Castro**, *Thesis Supervisor*.

Ungradgraduate Students

2022 **Piyush Arora**, *Project Co-Supervisor*, International Student.

Professional Activities

Organizer **eXplainable AI approaches for deep reinforcement learning**, *Workshop*, AAAI, 2024.

Organizer **eXplainable AI approaches for debugging and diagnosis**, *Workshop*, NeurIPS, 2020.

Reviewer **AISTATS**, 2022.

Volunteer **IJCAI**, 2020.

Soft skills

- Excellent experience with autonomous and remote work;
- Good experience on presenting research results and projects at conferences and international events;
- Good ability to adapt to multicultural environments;

- High predisposition to collaborations with different research groups (e.g., A.WA.RE Research Group, Sapienza NLP Group, Ro.Co.Co Lab, LaBRI, AIEA Lab);

Languages

Italian	Native
English	B2

Referees

Dr. Roberto Capobianco
Senior Research Scientist (Sony AI)
External Professor, Department of Computer, Control and Management Engineering
Sapienza University of Rome
✉ capobianco@diag.uniroma1.it

Dr. Leilani Gilpin
Assistant Professor, Department of Computer Science and Engineering
UC Santa Cruz
✉ lgilpin@ucsc.edu

Dr. Daniele Nardi
Full Professor, Department of Computer, Control and Management Engineering
Sapienza University of Rome
✉ nardi@diag.uniroma1.it